

SCHOOL OF COMPUTER SCIENCE AND IT YBN UNIVERSITY RANCHI

JAVA PROGRAMMING ASSIGNMENT 18

1. Basic Encapsulation

- **Description:** Create a class `Person` with private attributes `name` and `age`. Use getter and setter methods to access these fields.
- **Example:** Set a person's name to "John" and age to 30, then retrieve and display these values.
- **Test Cases:**
 - **Test 1:** Set `name` to "Alice", retrieve and verify it.
 - **Test 2:** Set `age` to 25, retrieve and verify it.

2. Data Validation in Setters

- **Description:** Enhance the `Person` class by adding validation to the setter methods to ensure that the age is non-negative and the name is not null or empty.
- **Example:** Try setting a negative age and catch the error.
- **Test Cases:**
 - **Test 1:** Attempt to set `age` to -5, expect error handling.
 - **Test 2:** Try setting `name` to an empty string, expect error handling.

3. Encapsulation with Calculations

- **Description:** Add a method to calculate the retirement year based on the age and current year. Assume retirement age is 65.
- **Example:** If a person is 30 years old in 2023, the retirement year should be calculated as 2058.
- **Test Cases:**
 - Input: `age = 30, currentYear = 2023`
 - Output: `retirementYear = 2058`

4. Immutable Classes

- **Description:** Create an immutable class `ImmutablePerson` where once a person object is created, its name and age cannot be changed.
- **Example:** Create an object with "Jane", 45, and any attempt to change these should be restricted.
- **Test Cases:**
 - Create object, attempt to modify `name` or `age`, ensure no changes are allowed.

5. Encapsulation in a Complex System

- **Description:** Design a class `Employee` that extends `Person`, adding private attributes like `employeeId` and `department`. Ensure all principles of encapsulation are followed, including appropriate getter and setter methods with validations.
- **Example:** Set `employeeId` to 1001 and `department` to "Engineering", and retrieve to display.
- **Test Cases:**
 - **Test 1:** Successfully set and get `employeeId`.
 - **Test 2:** Set `department` to "HR", then retrieve and verify.